

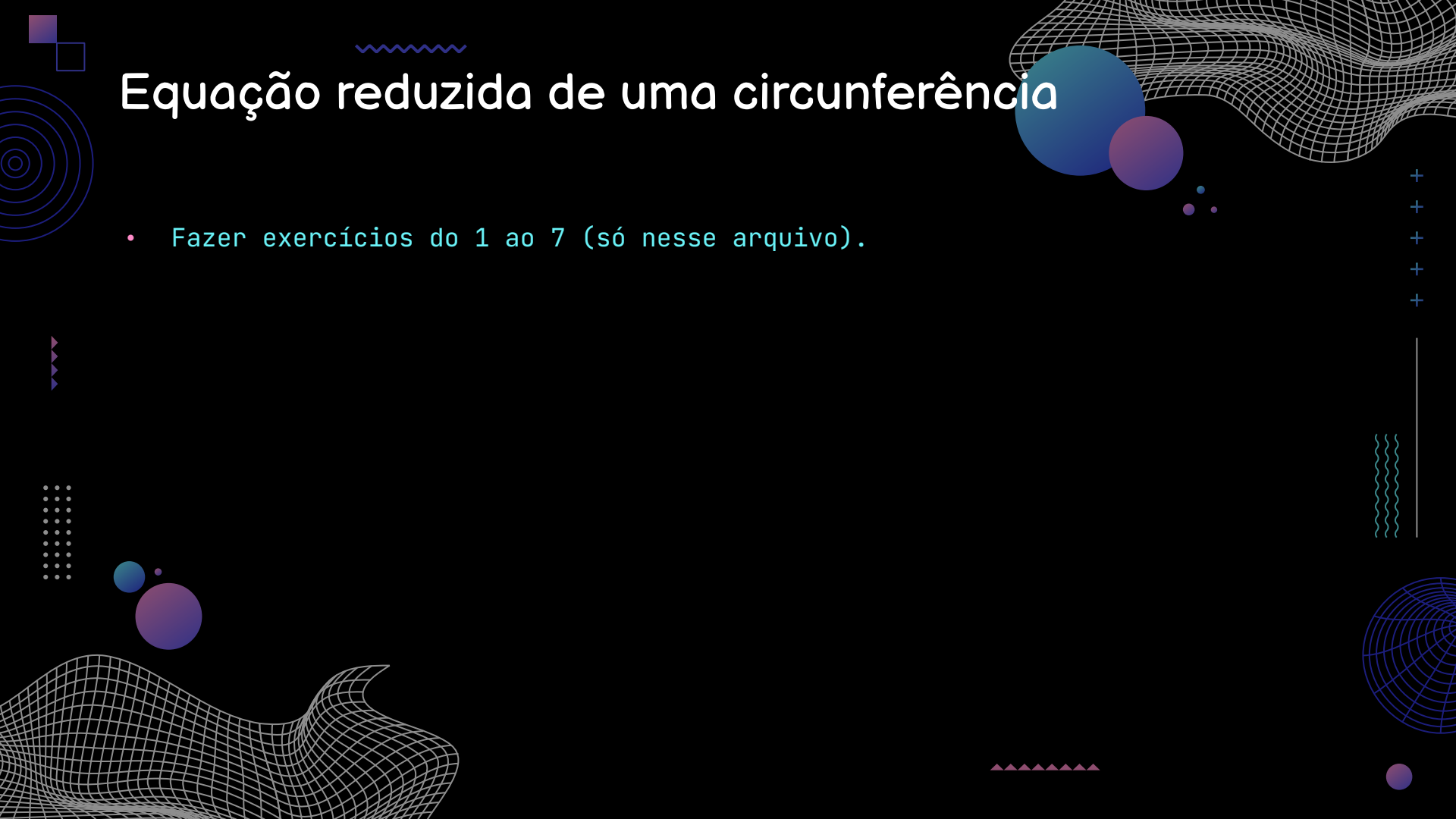


3ª Série

Matemática

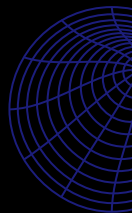
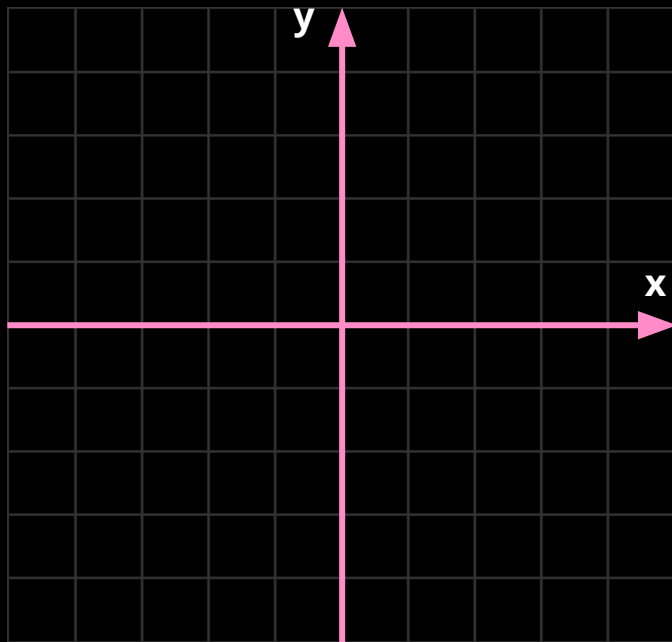
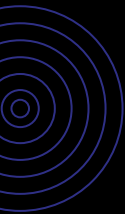
Aula 06 - 2º Bim

Prof. Diogo Pelaes

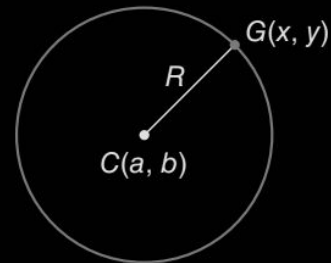


Equação reduzida de uma circunferência

- Fazer exercícios do 1 ao 7 (só nesse arquivo).



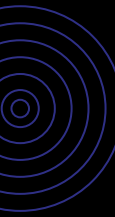
Equação reduzida de uma circunferência



$$(x - a)^2 + (y - b)^2 = R^2$$



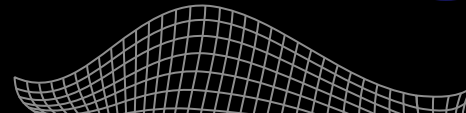
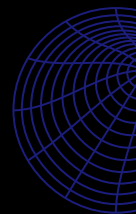
Exercício Resolvido



Obtenha a equação reduzida da circunferência de centro C e raio R em cada um dos seguintes casos:

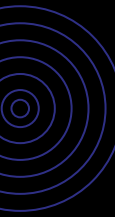
a. $C(4, 5)$ e $R = 7$

b. $C(0, 2)$ e $R = \sqrt{7}$



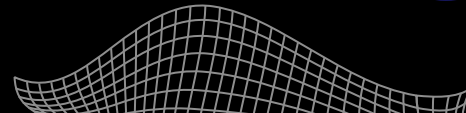
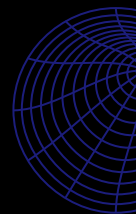


Exercício Resolvido



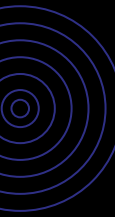
Determine o centro e o raio da circunferência que tem por equação:

$$(x + 2)^2 + y^2 = \frac{16}{9}$$

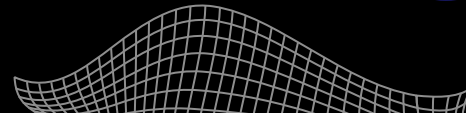
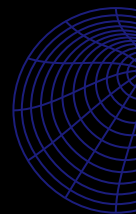
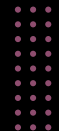




Exercício Resolvido

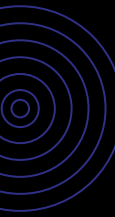


A equação $(x - 2)^2 + (y - 3)^2 = k$ representa uma circunferência para quais valores reais de k ?

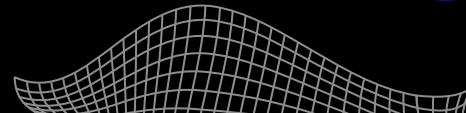
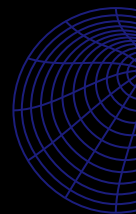




Exercício Resolvido

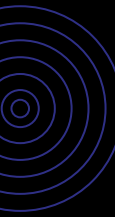


Qual é o conjunto de pontos (x, y) do plano cartesiano tal que $(x - 2)^2 + (y - 3)^2 = 0$?

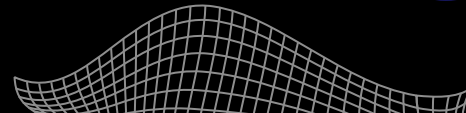
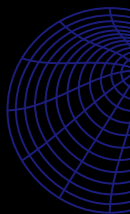




Exercício Resolvido



Qual é o conjunto dos pontos (x, y) do plano cartesiano tal que $(x - 2)^2 + (y - 3)^2 = -16$?





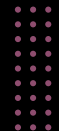
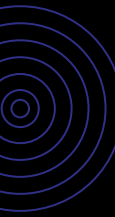
1. Determine a equação reduzida da circunferência de centro C e raio R nos seguintes casos:

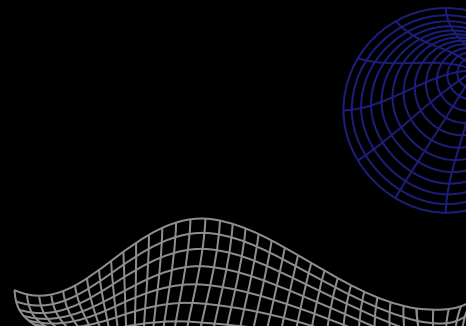
a. $C(4, 6)$ e $R = 10$

b. $C(7, 0)$ e $R = \sqrt{3}$

c. $C(0, -1)$ e $R = \frac{2}{3}$

d. $C(0, 0)$ e $R = 4$





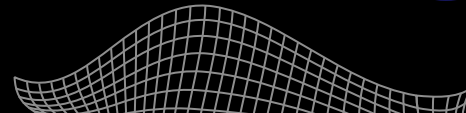
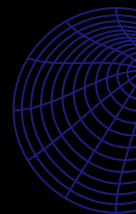


2. Obtenha o centro C e o raio R da circunferência de equação:

a. $(x - 6)^2 + (y - 2)^2 = 49$

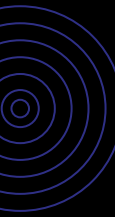
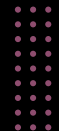
b. $(x - 4)^2 + y^2 = 5$

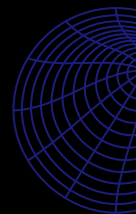
c. $(x + 1)^2 + (y + 2)^2 = \frac{16}{25}$



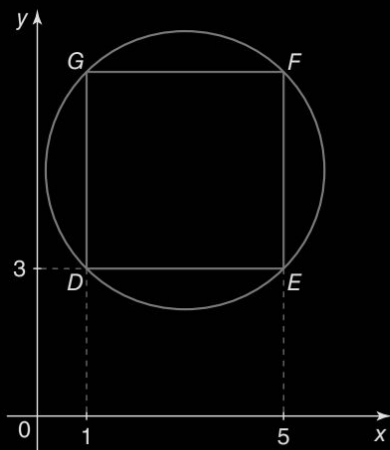


3. Em relação à circunferência λ de equação $(x - 2)^2 + (y + 6)^2 = 25$, responda aos itens a seguir.

- 
- a. Qual dos pontos $A(1, 4)$ e $B(6, -3)$ pertence a λ ?
- b. Para que valores reais de k o ponto $P(-1, k)$ pertence a λ ?
- 

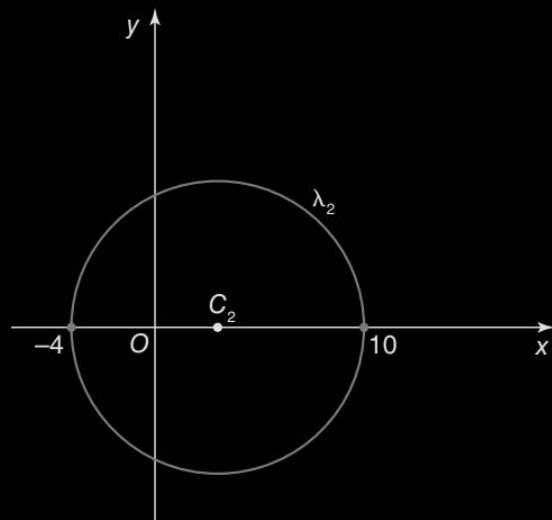
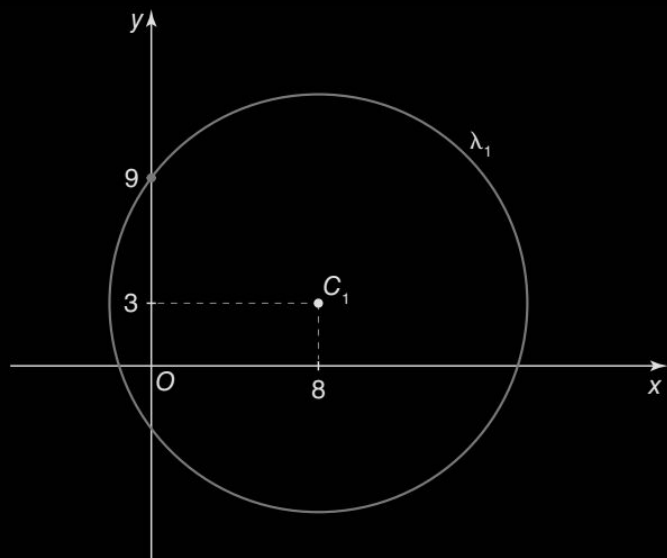


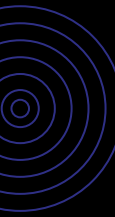
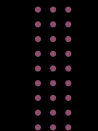
4. O gráfico a seguir mostra uma circunferência circunscrita a um quadrado $DEFG$, em que $D(1, 3)$ e $E(5, 3)$.

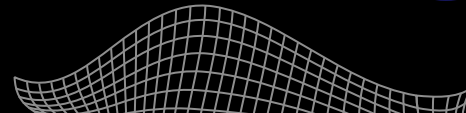
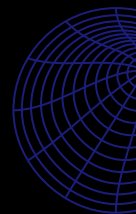


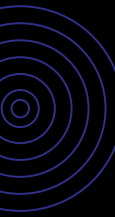
- Determine os pontos F e G .
- Obtenha a equação reduzida dessa circunferência.
- Determine o ponto P de abscissa máxima pertencente a essa circunferência.
- Determine o ponto Q de ordenada mínima pertencente a essa circunferência.

5. As circunferências λ_1 e λ_2 têm centros C_1 e C_2 , respectivamente. Obtenha a equação reduzida de cada uma delas.

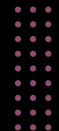


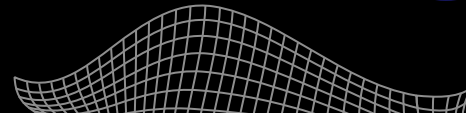
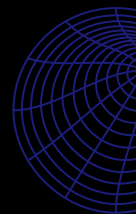
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6. Uma circunferência λ passa pelo ponto $P(6, -1)$, seu centro pertence à reta bissetriz dos quadrantes ímpares e seu raio mede 5 unidades. Obtenha a equação reduzida de λ .
- 





7. Considere a equação $(x - 5)^2 + (y + 1)^2 = k - 4$ nas variáveis x e y . Determine o(s) valor(es) real(is) de k de modo que essa equação represente:

- a. uma circunferência;
 - b. um ponto;
 - c. o conjunto vazio.
- 



Gabarito

1. a. $(x-4)^2 + (y-6)^2 = 100$

b. $(x-7)^2 + y^2 = 3$

c. $x^2 + (y+1)^2 = \frac{4}{9}$

d. $x^2 + y^2 = 16$

2. a. $C(6, 2)$ e $R=7$

b. $C(4, 0)$ e $R=\sqrt{5}$

c. $C(-1, -2)$ e $R=\frac{4}{5}$

3. a. A não pertence a λ , e B pertence a λ .

b. $k=-2$ ou $k=-10$

4. a. $F(5, 7)$ e $G(1, 7)$

b. $(x-3)^2 + (y-5)^2 = 8$

c. $P(3 + 2\sqrt{2}, 5)$

d. $Q(3, 5 - 2\sqrt{2})$

5. $\lambda_1: (x-8)^2 + (y-3)^2 = 100$ e $\lambda_2: (x-3)^2 + y^2 = 49$

6. $(x-2)^2 + (y-2)^2 = 25$ ou

$(x-3)^2 + (y-3)^2 = 25$

7. a. $k > 4$

b. $k = 4$

c. $k < 4$